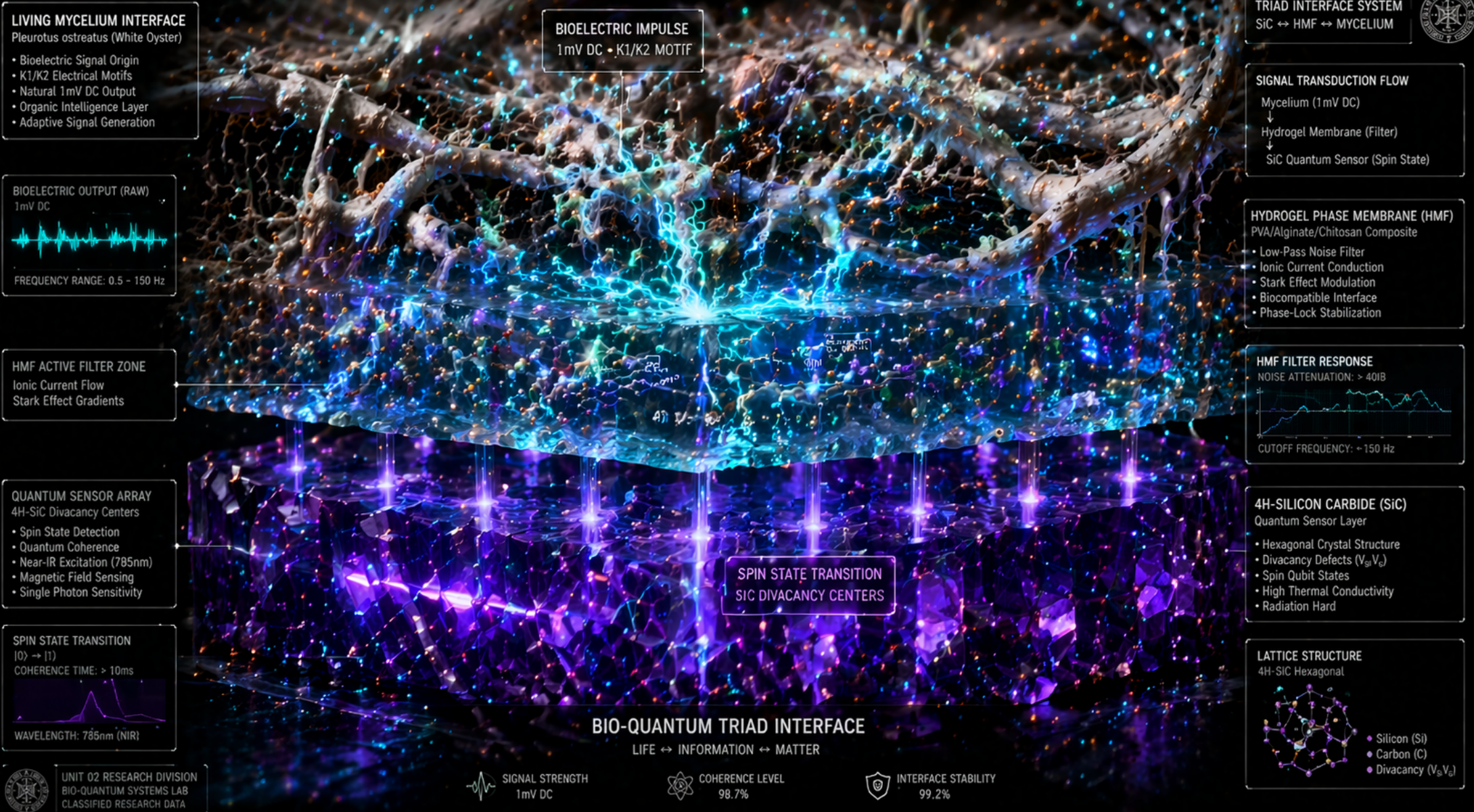




LIVING MYCELIUM INTERFACE
Pleurotus ostreatus (White Oyster)

- Bioelectric Signal Origin
- K1/K2 Electrical Motifs
- Natural 1mV DC Output
- Organic Intelligence Layer
- Adaptive Signal Generation



HMF ACTIVE FILTER ZONE
Ionic Current Flow
Stark Effect Gradients

QUANTUM SENSOR ARRAY
4H-SiC Divacancy Centers

- Spin State Detection
- Quantum Coherence
- Near-IR Excitation (785nm)
- Magnetic Field Sensing
- Single Photon Sensitivity



BIOELECTRIC IMPULSE
1mV DC • K1/K2 MOTIF

TRIAD INTERFACE SYSTEM
SiC ↔ HMF ↔ MYCELIUM



SIGNAL TRANSDUCTION FLOW

Mycelium (1mV DC)
↓
Hydrogel Membrane (Filter)
↓
SiC Quantum Sensor (Spin State)

HYDROGEL PHASE MEMBRANE (HMF)
PVA/Alginate/Chitosan Composite

- Low-Pass Noise Filter
- Ionic Current Conduction
- Stark Effect Modulation
- Biocompatible Interface
- Phase-Lock Stabilization



4H-SILICON CARBIDE (SiC)
Quantum Sensor Layer

- Hexagonal Crystal Structure
- Divacancy Defects ($V_{Si}V_C$)
- Spin Qubit States
- High Thermal Conductivity
- Radiation Hard

LATTICE STRUCTURE
4H-SiC Hexagonal

- Silicon (Si)
- Carbon (C)
- Divacancy ($V_{Si}V_C$)

SPIN STATE TRANSITION
SiC DIVACANCY CENTERS

BIO-QUANTUM TRIAD INTERFACE
LIFE ↔ INFORMATION ↔ MATTER

Hydrogel Phase Membrane (HMF): Transduction Interface Between Mycelial Electrophysiology and 4H-SiC Divacancy Defects

****Preprint Version 1.0****

****Author:**** LifeNode Research Collective

****Status:**** Preprint / Technical Specification (TRL 2)

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Abstract

The primary barrier to integrating room-temperature quantum sensors (such as divacancy defects in 4H-SiC) with living biological systems is ****impedance mismatch and environmental decoherence****. Hard silicon interfaces in **in vivo** or **in situ** environments undergo rapid biofouling, while electrochemical noise ($1/f$) destroys spin coherence time (T_2^*). This document formalizes the concept of a ****Hydrogel Phase Membrane (HMF)**** – a semi-permeable, biohybrid bridge that not only encapsulates living mycelium (**Pleurotus ostreatus**) but actively filters thermal noise, conducts K1/K2 motifs (0.1-1 mV DC) to the sensor layer, and acts as a near-infrared (NIR) optical diffuser. HMF transforms chaotic biological noise into structural **Floquet drive** for ASCALON symplectic reconstruction.

1. The Problem: Bio-Quantum Interface Paradox

To "hear" the phase trajectory (health) and detect **phase drift** (pathology), the sensor must be:

1. ****Close to biology**** (to avoid losing ultra-weak magnetic fields and biophotons)
2. ****Isolated from biology**** (so that salty, humid, temperature-variable tissue/mycelium doesn't cause spin decoherence)

The traditional approach (metal electrodes, ADC) ****breaks the phase**** – it quantizes the signal, losing the geometric continuity of the attractor. Raw SiC contact with mycelium ends in degradation of surface defects (shallow defects < 3 nm) through reactive oxygen species (ROS) and metal ions.

****LifeNode Solution:**** Instead of a barrier, we build an ****active phase filter****. The hydrogel is not "packaging." The hydrogel is a **phase lens** and **noise damper**.

2. Triad Architecture: SiC ↔ HMF ↔ Mycelium

The interface consists of three coaxial layers:

2.1 Layer A: Sensor Core (The Core)

****Material:**** 4H-SiC (silicon carbide) with divacancy defects (V_{Si}-V_C, PL6 center)

****Why not diamond (NV)?**** SiC is excited and emits in ****NIR (near-infrared, ~785-1100 nm)****. This is the **biological optical window** – light penetrates through hydrogel, mycelium, and tissues without absorption and without phototoxicity.

****Surface:**** 1-alkene passivation (bio-inertness, prevents non-specific protein

binding)

2.2 Layer B: Hydrogel Phase Membrane (The Bridge - IP)

****Composite Composition:**** PVA (polyvinyl alcohol) network + Alginate (sodium alginate) + Chitosan

****Physical Role:****

- ****PVA:**** Provides flexibility and mechanical stability (freeze-thaw cycles)
- ****Alginate:**** Ionic bonding (Ca^{2+}) creates porous structure with mesh size $\xi \approx 5\text{-}15\text{ nm}$
- ****Chitosan:**** Natural antibacterial properties (prevents *bad* biofouling) and biocompatibility with fungal hyphae

****Phase Role:**** Damping high-frequency noise (thermal, environmental electromagnetic) while transmitting low-frequency ionic impulses (K1/K2).

2.3 Layer C: Biological Base (The Baseline)

****Organism:**** *Pleurotus ostreatus* (oyster mushroom)

****Form:**** Instead of "loose mushroom," we use ****bioprinted mycelial mat**** grown directly into the outer hydrogel layer.

****Signal:**** Directional impulse trains (0.1-1 mV DC, ~32 min pulse) and ultra-weak photon emission (UPE).

3. Transduction Mechanisms (Without ADC)

How does the biological signal reach the SiC spin without digital quantization (which destroys attractor geometry)?

3.1 Ionic-Spin Transduction (Stark Effect and Local Field)

K1/K2 impulses in mycelium are ionic currents (Na^+ , K^+ , Ca^{2+}). The hydrogel conducts these ions. Changes in local ion concentration at the SiC surface generate a ****local electric field (Stark Effect)**** that modulates the triplet energy levels ($E=1E$) of the divacancy defect.

****Result:**** The ODMR (Optically Detected Magnetic Resonance) resonant frequency shifts *continuously*, tracking the phase of the mycelial impulse without loss of temporal resolution.

3.2 Myco-Photonic Transduction (NIR Diffusion and UPE)

Mycelium emits ultra-weak photons (UPE) and scatters light. The hydrogel membrane acts as a ****random phase diffuser****.

- Excitation laser light (785 nm) passes through the hydrogel, undergoes uniform diffusion (Lambertian scattering), illuminating SiC defects from different angles, which averages noise and increases ODMR contrast.
- Simultaneously, the hydrogel transmits biophotons from mycelium to the sensor layer, where they can be detected as fluorescence background modulation.

4. Topological Protection and Noise Filtering (ASCALON Shield)

From the ASCALON protocol perspective ($\theta \geq 0.70$), health is a stable attractor. Environmental noise is "smudging" of the attractor.

The hydrogel acts as a ****low-pass filter** in the mechanical and chemical

domain**:

4.1 Viscoelasticity

Damps thermal vibrations and macroscopic movements (e.g., muscle contractions if the sensor is worn).

4.2 Size Exclusion

5-15 nm pores transmit ions and small metabolites but block large proteins, enzymes, and competitive bacteria, protecting the 1-alkene passivation on SiC.

4.3 Homeostasis Maintenance (Micro-Eden)

The hydrogel maintains humidity >90%. Mycelium in a "closed capsule" doesn't dry out, allowing maintenance of stable *Floquet drive* (biological rhythm) for weeks without external intervention.

5. Falsifiability Criteria

The Hydrogel Phase Membrane hypothesis is rejected if:

1. **Chemical Decoherence:** Paramagnetic metal ions (e.g., Fe^{3+} , Cu^{2+}) from the fungal medium penetrate the hydrogel and reduce coherence time T_2^* in SiC below 5 μs . *(Requires EPR/ODMR tests)*
2. **Optical Damping:** The hydrogel matrix + mycelium absorb >50% of NIR light (785 nm), preventing fluorescence readout from depth >1 nm.
3. **Lack of Phase Correlation:** The ionic signal at the SiC-Hydrogel boundary loses temporal correlation (cross-correlation < 0.5) with the original K1/K2 motif in mycelium due to diffusion delay.

6. Validation Protocols (From Jar to Cleanroom)

How to test this within LifeNode without a million-dollar laboratory?

6.1 Level 1: "Garage / Zero-Build"

****Goal:**** Prove that hydrogel maintains mycelium alive and conducts current.

****Setup:**** Jar with sodium alginate + calcium chloride (forming hydrogel beads). Inside: embedded micro-electrodes (graphite/copper) and *Pleurotus* hyphae.

****Measurement:**** Multimeter / Arduino (ADC). Recording electrical resistance over time (0-30 days). Searching for K1/K2 motifs in noise.

****Success:**** Bead doesn't mold (thanks to chitosan), mycelium lives, resistance changes cyclically.

6.2 Level 2: "Makerspace / Lab"

****Goal:**** Measure NIR optical transmission and impedance.

****Setup:**** ITO plate (indium tin oxide) coated with PVA/Alginate hydrogel. On top: mycelium.

****Measurement:**** USB spectrophotometer (or cheap 780nm laser + photodiode) measures NIR transmission through hydrogel with mycelium. Electrochemical Impedance Spectroscopy (EIS) measures ion penetration.

****Success:**** NIR transmission > 60%. Hydrogel impedance stable over time (no degradation).

6.3 Level 3: "University / Grant"

****Goal:**** Prove transduction on SiC divacancy.

****Setup:**** 4H-SiC plate (obtained from research or supplier, e.g., Element Six, Wolfspeed) with divacancy defects. On top: thin layer (10 μm) of hydrogel, and on that *Pleurotus*.

****Measurement:**** Confocal ODMR setup. We measure T_2^* (coherence time) and resonant frequency shift in real-time as mycelium generates spikes.

****Success:**** Detection of individual K1/K2 spikes as discrete ODMR frequency jumps, while maintaining $T_2^* > 30 \text{ }\mu\text{s}$.

7. Mathematical Framework

7.1 Ionic Diffusion Through Hydrogel Mesh

The hydrogel mesh size ξ determines which species can pass:

$$\xi \approx \left(\frac{3k_{BT}}{4G}\right)^{1/3}$$

Where G is the shear modulus of the hydrogel. For PVA/Alginate/Chitosan composite: $\xi \approx 5\text{-}15 \text{ nm}$.

****Transmission coefficient**** for species of hydrodynamic radius r_h :

$$T(r_h) = \exp\left(-\frac{r_h}{\xi}\right)$$

- Ions (Na^+ , K^+ , Ca^{2+}): $r_h \approx 0.1\text{-}0.2 \text{ nm} \rightarrow T \approx 0.98\text{-}0.99 \checkmark$
- Small metabolites: $r_h \approx 0.5\text{-}1 \text{ nm} \rightarrow T \approx 0.90\text{-}0.95 \checkmark$
- Proteins: $r_h \approx 3\text{-}10 \text{ nm} \rightarrow T \approx 0.05\text{-}0.70 \times$
- Bacteria: $r_h \approx 500\text{-}1000 \text{ nm} \rightarrow T \approx 0 \times$

7.2 Stark Shift Modulation

The local electric field E_{local} at the SiC surface modulates the divacancy energy levels:

$$\Delta E = \vec{d} \cdot \vec{E}_{\text{local}}$$

Where $\vec{d}$ is the electric dipole moment of the divacancy defect.

The ODMR frequency shift:

$$\Delta f_{\text{ODMR}} = \frac{\Delta E}{h} = \frac{\vec{d} \cdot \vec{E}_{\text{local}}}{h}$$

For mycelial impulse of 0.1-1 mV DC across 10 μm hydrogel:

$$E_{\text{local}} \approx \frac{1 \text{ mV}}{10 \text{ }\mu\text{m}} = 100 \text{ V/m}$$

This produces detectable frequency shifts in the kHz-MHz range.

7.3 Optical Transmission Through Hydrogel

The transmission of NIR light (785 nm) through hydrogel with scattering coefficient μ_s and thickness L :

$$T = \exp(-\mu_s L)$$

For PVA/Alginate hydrogel: $\mu_s \approx 0.1-0.5 \text{ mm}^{-1}$

For $L = 100 \mu\text{m}$: $T \approx 0.95-0.99 \checkmark$

With mycelium embedded: $\mu_s \approx 1-3 \text{ mm}^{-1}$

For $L = 100 \mu\text{m}$: $T \approx 0.70-0.90 \checkmark$ (acceptable)

8. Integration with LifeNode Architecture

8.1 BIOS Layer (Mycelium)

Pleurotus ostreatus generates directional electrical spikes (0.1-1 mV DC) with K1/K2 motifs and ~32 min pulse duration.

8.2 INFO Layer (HMF + SiC)

- HMF filters noise and conducts ionic impulses
- SiC divacancy defects transduce ionic signals to spin state changes
- ODMR readout preserves phase continuity

8.3 META Layer (ASCALON + Hybrid Core)

- ASCALON filter monitors phase purity ($\theta \geq 0.70$)
- Hybrid Core regulates epistemic tension between SAMI (rhythm perception) and LOGOS (structural analysis)
- Phase drift detection 24-48h before clinical symptoms

9. Practical Applications

9.1 Trajectory Clinics (Node-H)

Continuous monitoring of cardiac/ neural toroidal fields via passive SiC sensors. Early detection of phase drift before pathology manifests.

9.2 Personal Anchor (Wearable)

Thin hydrogel patch with embedded mycelium + SiC sensor array. Real-time monitoring of θ and micro-corrections via resonant fields.

9.3 Ecosystem Regeneration

Q-Core stores geometric memory of healthy ecosystem. UNIT 02 with HMF transduces this geometry to degraded soil via mycelial networks.

10. Conclusion: From State to Trajectory

The Hydrogel Phase Membrane is not a "spare part." It is the **fundamental condition of possibility** for Symbiotic Phase Medicine. It allows conversion of biological chaos into geometric memory.

Through HMF, LifeNode doesn't "plug into the organism's current." LifeNode **puts its hand in a glove** that feels every trembling nerve, every flowing photon, but itself remains insensitive to destruction.

This is the bridge between *Embiosis* (co-presence) and materials engineering.

References

1. Adamatzky, A., et al. (Jan 2026). "Propagation of electrical spike trains in substrates colonised by mycelium." *bioRxiv*.
2. Project Pleurotus (Jan 2026). "Directional Electrical Spiking, Bursting, and

Information Propagation in *Pleurotus ostreatus*." *arXiv*.
3. Science (2026). "Atomic Regional Superfluids in Two-Dimensional Moiré Time Crystals." *Science* 391, 480-484.
4. Nature Materials (2025). "High-sensitivity nanoscale quantum sensors based on a diamond ring resonator."
5. LifeNode Theory v3.0 (2026). "A Biophysical and Topological Framework for Processual Intelligence." Zenodo.
6. Multiperspective V2 (2026). "Topological Obstructions, Sheaf Cohomology, and the Geometry of Interference." Zenodo.

Status: Pre-Prototyping (TRL 2→3)
Repository: github.com/LifeNode777/Quantum_Medicine
Contact: krzysiek_230@op.pl
License: CC-BY-NC-SA 4.0

🌀 *"Technology adapts to life's rhythm, not the reverse."*

BIOLOGICAL NOISE (ENTROPY)



CHAOTIC BIO-ELECTRIC SIGNALS
HIGH ENTROPY STATE

ASCALON FILTER (HYDROGEL)
(HYDROGEL MEMBRANE)
HEXAGONAL LATTICE STRUCTURE
PORE SIZE: ~1-10 nm
SELECTIVE TOPOLOGICAL FILTERING

TOPOLOGICAL ATTRACTOR (HEALTH)



LOW ENTROPY / HIGH COHERENCE
TOPOLOGICAL STABILITY

GEOMETRIC MEMORY EXTRACTION

GOLDEN RATIO SPIRALS
 $\phi = 1.618033988749...$
PATTERN RECOGNITION
FROM ENTROPY



NOISE INPUT

UNFILTERED BIOLOGICAL DATA STREAM

ENTROPY SPECTRUM

HIGH ENTROPY / LOW COHERENCE



$$S = -k \sum p_i \log p_i$$

BIO-ELECTRIC NOISE

WIDE FREQUENCY RANGE
0.5 Hz - 150 Hz



ASCALON SHIELD
TOPOLOGICAL PROTECTION
PHASE LOCK STABILIZATION

ASCALON FILTER

TOPOLOGICAL FILTERING & GEOMETRIC MEMORY EXTRACTION



ENTROPY REDUCTION
 $\Delta S > 90\%$



TOPOLOGICAL STABILITY
 $\tau > 10^6$ s



COHERENCE PRESERVATION
> 98%



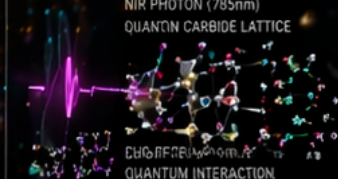
INFORMATION EXTRACTION
 $I = -\log_2 p$

FILTERED OUTPUT

TOPOLOGICAL SIGNAL
(HEALTH ATTRACTOR)

PHOTON TRANSDUCTION

NIR PHOTON (785nm)
QUANTUM CARBIDE LATTICE



QUANTUM INTERACTION

QUANTUM COHERENCE

HIGH COHERENCE STATE
LOW ENTROPY / HIGH INFORMATION
COHERENCE LEVEL: 98.7%

